Han, Zhixian

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Research Interests

computational cognitive neuroscience; biologically inspired neural networks; cognitive psychology; computational modeling of neural and behavioral data; neural decoding; large language models and AI agents.

Professional Experience

2026 – Present Postdoctoral Researcher, Purdue University

2025 – 2026 Limited-Term Lecturer and Postdoctoral Researcher, Purdue University

2025 Graduate Lecturer, Purdue University

2024 Consultant, Purdue Statistical Consulting Service (SCS), Purdue University

2019 – 2024 Graduate Teaching and Research Assistant, Purdue University

2018 – 2019 Graduate Research Assistant, Brown University

2015 – 2017 Undergraduate Research Assistant, University of Minnesota, Twin Cities

Education

2019 – 2025	Ph.D. in Psychology (Mathematical and Computational Cognitive Science), Purdue University
2021 – 2025	Joint M.S. in Statistics and Computer Science, Purdue University
2019 – 2021	M.S. in Psychology, Purdue University
2017 – 2019	M.S. in Physics, Brown University
2014 – 2017	B.S. in Physics and B.S. in Psychology, University of Minnesota, Twin Cities
2015	Summer Visiting Undergraduate Student, University of California, Berkeley
2012 – 2014	Undergraduate studies in Physics, Sichuan University (transferred)

Peer-reviewed Journal Articles

Yi-Fan Li, Elizabeth M. Edwards, **Zhixian Han**, and Anne Sereno. "Using neural networks to understand static and dynamic cues in facial expression recognition". In: *Vision Research* 244 (2026), pp. 1–13. DOI: 10.1016/j.visres.2026.108827.

Zhixian Han and Anne B. Sereno. "Exploring Neural Architectures for Simultaneously Recognizing Multiple Visual Attributes". In: *Scientific Reports* 14, 30036 (2024). DOI: 10.1038/s41598-024-80679-6.

Zhixian Han and Anne B. Sereno. "Understanding Cortical Streams From a Computational Perspective". In: *Journal of Cognitive Neuroscience* 36.12 (2024), pp. 2618–2626. DOI: 10.1162/jocn_a_02121.

Zhixian Han and Anne B. Sereno. "A Spatial Map: A Propitious Choice for Constraining the Binding Problem". In: *Frontiers in Computational Neuroscience* 18 (2024). DOI: 10.3389/fncom.2024.1397819.

Zhixian Han and Anne Sereno. "Identifying and Localizing Multiple Objects Using Artificial Ventral and Dorsal Cortical Visual Pathways". In: *Neural Computation* 35.2 (2023), pp. 249–275. DOI: 10.1162/neco_a_01559.

Zhixian Han and Anne Sereno. "Modeling the Ventral and Dorsal Cortical Visual Pathways Using Artificial Neural Networks". In: *Neural Computation* 34.1 (2022), pp. 138–171. DOI: 10.1162/neco_a_01456.

Conference Abstracts

Raiyana Z. Alam, **Zhixian Han**, Yi-Fan Li, and Anne B. Sereno. "Analyzing the Role of Spatial Attention in the Ventral Stream Using fMRI". In: *Indiana University Medical Student Program for Research and Scholarship Symposium*. Indianapolis, IN, July 23, 2026.

Zhixian Han and Anne Sereno. "Cross-Stream Contrastive Learning Inspired by Feature-Biased Visual Processing". In: *Computational and Mathematical Models in Vision (MODVIS)*. to appear. 2026.

Anne Sereno and **Zhixian Han**. "Insights into the reasons for segregated processing and the privileged role of space in binding, from a computational perspective". In: *Journal of Vision*. to appear. 2026.

Anne Sereno, **Zhixian Han**, and Elizabeth M. Frazier. "A Computational Perspective on Segregated Processing and the Privileged Role of Space in Binding". In: *Computational and Mathematical Models in Vision (MODVIS)*. 2025. URL: https://docs.lib.purdue.edu/cgi/viewcontent.cgi?article=1248&context=modvis.

Zhixian Han, Vlada Volyanskaya, and Anne B. Sereno. "A Biologically Inspired Framework for Contrastive Learning of Visual Representations: BioCLR". In: *Journal of Vision*. 2024. DOI: 10.1167/jov.24.10.771.

Zhixian Han and Anne Sereno. "Is It Always Computationally Advantageous to Use Segregated Pathways to Process Different Visual Stimulus Attributes Separately?" In: *Journal of Vision*. 2023. DOI: 10.1167/jov.23.9.5020.

Anne B. Sereno and **Zhixian Han**. "Independence, Not Interactions: What Simulations Suggest about Ventral and Dorsal Pathways". In: *Journal of Vision*. 2023. DOI: 10.1167/jov.23.9.4606.

Zhixian Han and Anne Sereno. "Constraining the Binding Problem Using Maps". In: *Computational and Mathematical Models in Vision (MODVIS)*. 2023. URL: https://docs.lib.purdue.edu/modvis/2023/session03/2/.

Zhixian Han and Anne Sereno. "Identifying and Localizing Multiple Objects Using Artificial Ventral and Dorsal Visual Cortical Pathways". In: *Computational and Mathematical Models in Vision (MODVIS)*. 2022. URL: https://docs.lib.purdue.edu/modvis/2022/session01/3/.

Manuscripts Under Review

Zhixian Han, Yi-Fan Li, Elizabeth M. Edwards, and Anne B. Sereno. "Cross-Stream Self-Supervised Learning Inspired by Feature-Biased Visual Processing in the Brain." Submitted to *Transactions on Machine Learning Research*. Preprint under screening at bioRxiv.

Working Papers

Zhixian Han, Prasiddhi Shivakumaran, and Anne B. Sereno. "Induced Emotion and Word Recognition: A Diffusion Model Analysis." Manuscript in preparation.

Computational and Quantitative Skills

Computational Modeling: Artificial Neural Networks, Self-Supervised Learning, Bayesian Modeling, Simulation-Based Inference, Mixed-Effects Models, GPU-Accelerated Model Training
Neural Data Analysis: fMRI Preprocessing, GLM Analysis, Single-Trial Beta Estimation, Multivariate Pattern Analysis, Neural Decoding
Quantitative Training: Probability, Statistics, Machine Learning, Statistical Physics
Programming: Python (PyTorch, TensorFlow), R, MATLAB, C++, SQL

Research Experience

Attentional Modulation of Shape and Location Representations in Human Visual Cortex: An fMRI Study May 2026 – Present

Postdoctoral Researcher, Purdue University; Advisor: Dr. Anne Sereno

- Investigate how shape attention and spatial attention modulate neural representations of object shape and spatial location in the human visual cortex.
- Analyze fMRI data from shape-attention, location-attention, and fixation tasks involving different shapes presented at different spatial locations.
- Implement fMRI preprocessing, single-trial beta estimation, and multivariate pattern decoding pipelines to measure shape and location information across lower- and higher-level visual areas in the ventral and dorsal cortical pathways.
- Compare decoding performance across shape-attention, location-attention, and fixation conditions to determine how attentional demands affect visual representations across different visual areas.

Cross-Stream Self-Supervised Learning of Visual Representations September 2025 – Present

Postdoctoral Researcher, Purdue University; Advisor: Dr. Anne Sereno

- Develop a contrastive self-supervised learning framework for cross-stream visual representation learning, enabling interactions between specialized visual streams.
- Implement teacher–student distillation to incorporate learning signals for visual attributes into self-supervised representation learning.
- Analyze how interactions between specialized streams trained on different visual attributes shape learned visual representations.

Modeling Emotional Effects on Lexical Decision Performance April 2026 – Present

Postdoctoral Researcher, Purdue University; Advisor: Dr. Anne Sereno

- Investigate how anxiety and serenity inductions influence the cognitive processes underlying lexical decisions for positively and negatively valenced words, as well as nonwords.
- Develop a hierarchical drift-diffusion and attentional-lapse model with separate word and nonword evidence-accumulation rates, shared decision boundaries and starting-point bias, lexical-status-dependent nondecision and invalid-response processes, and participant-specific latent cognitive parameters.
- Apply simulation-based inference, parameter recovery, and posterior predictive checks to determine whether emotion-related behavioral differences arise from changes in evidence accumulation, decision caution, starting-point bias, nondecision processing, or attentional lapses.

Recognition-Action Segregation in Pretrained Large Language Model Agents

September 2026 – Present

Postdoctoral Researcher, Purdue University

- Develop a biologically inspired dual-pathway architecture built on pretrained large language models. Separate recognition (world-model) computation from action (policy) computation to investigate functional specialization in LLM-based AI agents.
- Fine-tune pretrained LLMs to form a Recognition pathway that predicts the next environmental state from task context, interaction history, current state, and action, and an Action pathway that selects the next action from task context, history, and current state.
- Compare segregated and integrated architectures under matched model-capacity and training conditions.
- Design a joint recognition-action task in which the model receives the current state s_t and produces s_{t+1} through sequential Action and Recognition pathway processing, without receiving the true action a_t. Test whether this specialized two-pathway architecture outperforms a parameter-matched integrated model on the joint task.

Understanding the Visual Cortical Pathways from a Computational Perspective

Graduate Researcher, Purdue University; Advisor: Dr. Anne Sereno September 2019 – August 2025

- Developed artificial neural network models of ventral and dorsal cortical pathways to investigate how architectural specialization shapes neural computation, learned representations, generalization, and visual binding.
- Designed experiments comparing segregated and integrated architectures for processing and binding visual attributes such as identity, location, luminance, and orientation.
- Demonstrated how specialized neural pathways support complementary neural representations, efficient learning, and generalization.
- Produced multiple first-author peer-reviewed publications in computational neuroscience and neural computation.

Neural Decoding of Word Spectrograms from Intracortical Recordings May 2018 – May 2019

Graduate Researcher, Brown University; Advisor: Dr. Wilson Truccolo

- Used long short-term memory (LSTM) models to decode word spectrograms from neural signals recorded via intracortical microelectrode arrays in auditory superior temporal cortex.
- Implemented phase retrieval algorithms to reconstruct acoustic signals from decoded spectrograms.
- Combined neural decoding and signal reconstruction methods to recover word-level acoustic information from neural recordings.

Graph Neural Networks for Scientific Prediction August 2021 – April 2022

Graduate Data Science Researcher, The Data Mine, Purdue University; Corporate Partner: ATOM

- Collaborated with Accelerating Therapeutics for Opportunities in Medicine (ATOM) on machine learning approaches to opioid receptor pIC50 prediction.
- Implemented graph convolutional neural networks and random forest models in Python to predict molecular properties from scientific data.
- Developed experience applying machine learning to interdisciplinary scientific problems.

Bayesian Theory and Visual Perception of Size and Depth September 2016 – May 2017

Undergraduate Researcher, University of Minnesota; Advisor: Dr. Daniel Kersten

- Modeled visual perception of object size and depth as a Bayesian inference problem.
- Used Markov chain Monte Carlo (MCMC) sampling methods to approximate posterior distributions.
- Applied probabilistic modeling to study perceptual inference in human vision.

Characterizing Fluorescent Properties of mEos2 Proteins for Super-Resolution Microscopy September 2015 – May 2017

Undergraduate Research Assistant, University of Minnesota; Advisor: Dr. Elias Puchner

- Characterized the fluorescent properties of mEos2 proteins in yeast cells for super-resolution microscopy.
- Experimentally determined the optimal excitation laser power for imaging mEos2 proteins.
- Optimized imaging conditions to improve signal-to-background ratio and obtain high-quality super-resolution images.

Ad Hoc Reviewer

Neural Networks, Scientific Reports, Vision Research, Trends in Cognitive Sciences

Teaching Experience

Project Teaching Assistant – Neuromatch Academy July 2026

Computational Neuroscience Summer Course 2026 (Synchronous Online)

– Guided and supported eight project groups of five to seven students in developing research questions, conducting literature reviews, writing Python code, and preparing abstracts and project presentations.

Limited-term Lecturer – PSY 39200CPAI Intro to Computational Psychology and AI - Purdue University

August 2025 - May 2026

– Designed a new upper-level undergraduate course in computational psychology and AI.
– Created tutorials with theories and programming exercises.
– Prepared lectures to introduce different topics to students.
– Guided student discussions and course projects.

Limited-term Lecturer – PSY 20100 Intro to Statistics in Psychology - Purdue University

August 2025 - May 2026

– Created tutorials with theories and calculation exercises.
– Prepared lectures to introduce different statistics topics to students.

Graduate Lecturer – PSY 32400 Intro to Cognitive Neuroscience (Asynchronous Online) - Purdue University

June 2025 - July 2025

– Prepared lectures to introduce the neurobiological foundations of cognition and the computational principles underlying them.

Graduate Lecturer – PSY 51300 Intro to Computational Cognitive Neuroscience - Purdue University

January 2025 - May 2025

– Redesigned and taught an upper-level course covering computational cognitive modeling, reinforcement learning, convolutional neural networks, transformers, natural language processing, large language models, active inference, and NeuroAI.
– Created tutorials with theories and programming exercises.
– Prepared lectures to introduce different topics to students.
– Guided student discussions and course projects.

Lead Teaching Assistant – Neuromatch Academy July 2021

Computational Neuroscience Summer Course 2021 (Synchronous Online)

– Guided small groups of students in all aspects of live online learning. This included guiding students in completion of code-based tutorials, guiding students in contextualizing the problem sets, and guiding students in developing peer-programming and self-learning skills.
– Guided students in their final project.
– Managed and provided support for 7-8 junior teaching assistants.

Teaching Assistant – Purdue University August 2019 - December 2024

- PSY 30500 Understanding and Analyzing Psychological Data (Fall 2024)
- PSY 20100 Introduction to Statistics in Psychology (Spring 2020, Fall 2021, Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024)
- PSY 20000 Introduction to Cognitive Psychology (Fall 2019, Spring 2021)
- PSY 48400 The Psychology of Consciousness (Fall 2020)
- STAT 51400 Design of Experiments (Summer 2022)

Work Experience

Consultant for the Purdue Statistical Consulting Service (SCS) January 2024 - December 2024

- Experimental Design and Data Analysis Consulting: This service assists with all phases of research projects, including: proposal preparation, design of studies, survey design, data input strategies, data import/export, analysis of data, interpretation of results, presentation of results, other statistics or probability problems.
- Statistical Software Consulting: This service provides assistance with running a wide variety of statistical computing programs.

Training / Workshops

Unlocking New Horizons in Cognitive Modeling with Simulation-Based Inference, Ohio State University 2026

Neuromatch Academy Summer Course in NeuroAI, Online 2024

Neuromatch Academy Summer Course in Deep Learning, Online 2021

Neuromatch Academy Summer Course in Computational Neuroscience, Online 2020

Berkeley Summer Course in Mining and Modeling Neuroscience Data, UC Berkeley 2018

Service

Purdue Han Culture Association, Purdue University
Treasurer (May 2023 – Aug 2024); President (Aug 2024 – Oct 2024)

- Organized cultural events promoting Chinese culture (e.g., Mid-Autumn Festival, Lantern Festival, Chinese Culture Night Market, Master Chef competition)
- Managed budget and secured funding for events
- Facilitated community engagement and cross-cultural activities

Awards

Postdoctoral Supplemental Travel Award, 2026

Compton Graduate Student Research Travel Award, 2024

Purdue Graduate Student Government (PGSG) First-Tier Travel Grant, 2024

Purdue Graduate Student Government (PGSG) Third-Tier Travel Grant, 2023

Yan Jici Physics Scholarship, Institute of Physics Chinese Academy of Sciences, 2014

First-Class Subject Scholarship, Sichuan University, 2014

Third-Class General Scholarship, Sichuan University, 2013

Outstanding Student, Sichuan University, 2013

References

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